# FW-mid pro Omnidirectional Drive-by-wire Chassis

**User manual V2.0.3**

# CONTENTS

# 1. Foreword

(1) Thank you for purchasing our product, this user manual is applicable to FW-mid pro Omnidirectional Drive-by-wire Chassis (hereby referred to as "FW-mid pro").

(2) Before use, please carefully read this user manual and attentions, and correctly use strictly in accordance with this manual.

(3) For the loses caused by serious violation of this user manual, we undertake no responsibilities.

(4) Please well keep this manual for user reference during your operation.

(5) Professionals are required for commissioning, connection and installation of the chassis equipment to avoid irretrievable loses.

(6) DO NOT install, remove or replace equipment lines with electricity. If it is necessary to commission this product with electricity, please select the special commissioning tools with good insulation.

(7) Please use this product under the conditions allowed by laws and regulations, so that the public property or life safety will not be affected.

(8) We will irregularly update this product, the contents of update will be added into the new manual without notification.

(9) This manual may contain the contents which are not correct in technology or which do not comply with the operation. In case of problems which cannot be solved during use of this manual, please contact with the customer service or technical department of us.

(10) As for the contents of this manual, we will try our best to ensure that they are correct and accurate. In case of any improper or incorrect contents, please contact us for confirmation, thank you!

# Safety Information

The information herein does not include how to design, install or operate a complete robot, nor the peripheral equipment which may affect the safety of this complete system. The design and use of the complete system comply with the safety requirements formulated in the national standards and specifications. The integrators and end customers of FW-mid pro are responsible for being sure to comply with practical laws and regulations of relevant countries to ensure that the application of the complete robot will not cause any major danger. These include but are not limited to the following:

## ■ Effectiveness and responsibilities:

- A risk evaluation shall be conducted to the complete robot system. All the additional safety equipment of other machineries defined by risk evaluation shall be connected. It shall be ensured that, the design and installation of the peripheral equipment of the whole robot system, including software and hardware system, are correct.
- This robot is not equipped with relevant safety functions that a complete autonomously moveable robot shall have, including but not limited to automatic collision avoidance, fall prevention and alarm for creature approaching, etc. For relevant functions, the integrators and end customers are required to conduct safety evaluation in accordance with relevant regulations and feasible laws and regulations to ensure that the developed robot has no any major danger or potential safety hazard during actual application.
- Collecting all the documents of technical files: Including risk evaluation and this manual. Before operation and use of equipment, the existing safety risks may be known.

## ■ Environments:

- For first use, please carefully read this manual to understand the basic contents and operation specifications.
- For remote operation, please select the areas which are relevantly open. This chassis is not equipped with any sensor for automatic obstacle avoidance.

- This chassis shall be used under the temperature of -20°C~60°C.
- The chassis is not customized for IP protection grade, the IP protection grade of this chassis is IP33.

## Inspection:

- Inspecting whether the battery of chassis is fully charged.
- Ensuring that the chassis has no abnormality.
- Inspecting whether the battery of remote controller is fully charged.

## Operation:

- When using the remote control for debugging, please make sure the remote control is turned on and the vehicle can receive the remote control commands.
- Ensuring that operation is conducted in a relatively open place. And remote control shall be conducted with sight distance.
- FW-mid pro The maximum load is 80KG, during use, it shall be ensured that the effective load does not exceed 80KG.
- When the device reports low battery, please charge it in time. When the device malfunctions, please stop using it immediately to avoid secondary injuries.
- When the device malfunctions, please contact the relevant technical personnel and do not handle it without authorization.
- Please use the equipment in the environment which meets the IP protection grade requirements of the equipment.
- When charging, please ensure that the environment temperature is above 0°C.

## Maintenance:

- If the tire wear is severe, please replace it in time.
- If the battery is not used for a long time, it needs to be periodically charged every month when it is fully charged.
- The battery needs to be charged at least once a month.

# 2. Introduction

FW-mid pro is an omnidirectional drive-by-wire robotics chassis. It adopts swerve steering motor drive structure, which is a four-wheel independent drive and independent steering to realize precise and flexible control of chassis. FW-mid pro has an excellent traveling ability and off-road performance. With multiple moving modes, it can be used in indoor and outdoor applications. Meanwhile it combining the advantages of Ackermann steering and differential drive form. Compared with these drive form, the wearing of tire is lighter, the noise of driving is lower and vehicle operating more stable. By the modules and navigation systems of LiDAR, GPS and manipulator, etc., this chassis is widely used in inspection, patrol, detection, transportation, logistics, scientific research and various new applications and explorations requiring for mobile chassis.

## 2.1. Product list

After delivery, please carefully confirm the product list:

Chassis *1

Remote controller *1

Charger (48V) *1

Product manual *1

## 2.2. Performance parameters

Table 2 - 1 FW-mid pro performance parameter table

| Parameter type | Performance | Parameter |
|---|---|---|
| Structural size and weight | Dimensions(W*D*L) | 725*515*440mm |
| | Weight | 70kg |
| | Drive | Swerve Steering and Motor Drive |
| | Suspension | Independent suspension |
| | Material | Aluminium Alloy |
| | Ground clearance | Suspension 90mm/chassis bottom 110mm |
| | Wheelbase | 400mm |
| | Wheel track | 420mm |
| | Tire type/diameter | 240mm |
| Basic configuration | Driving motor | 350W*4 wheel hub motor |
| | Battery type | 48V/20AH lithium battery/BMS management system |
| | Charging time | 2-3h |
| | Charging method | 48V/10A manual charging adapter |
| | External power supply | 48V/10A-24V/15A-12V/15A |
| | Braking mode | Motor brake |
| | Parking method | Motor parking |
| Safety measures | Emergency stop button | √ |
| | Command check | √ |
| | Heartbeat protection | √ |
| | Current protection | √ |
| | Temperature protection | √ |
| VCU configuration | Dominant frequency | 168MHz |
| | Hardware floating point acceleration | √ |
| | Kinematic analysis | √ |
| | Communication interface | CAN interface |
| | Communication protocol | CAN 2.0B |
| Performance parameters | Remote control distance | 100m |
| | Vertical load (level road) | 80kg |
| | Speed | 0-5.4km/h |
| | Mileage | 30km |
| | Minimum turning radius | 0m |
| | Wading depth | 100mm |
| | Maximum climbing angle | 15° (full load) |
| | Crossing width | 120mm (full load) |
| | Obstacle surmounting height | 40mm (full load) |
| | Protection level | IP33 |
| | Operating temperature | -20°C~50°C |

# 3. Product Presentation

This section provides a basic introduction to the FW-mid pro mobile robot chassis, aiming to give users and developers a fundamental understanding of the FW-mid pro chassis. The FW-mid pro chassis adopts a modular and automotive-grade design approach, featuring independent omni-directional steering, allowing all four wheels to achieve 360° rotation in place. It supports lateral movement, Ackermann steering mode, diagonal movement mode, and spinning mode, providing agile maneuverability. The tires are durable and resistant to wear, enabling precise control over the chassis steering. With four independent wheel drives, the FW-mid pro chassis delivers powerful and ample power, excelling in climbing performance and minimizing slippage. It adapts well to various road conditions and utilizes the standard CAN communication protocol, offering multiple application modules and universal standard interfaces. This allows users to quickly integrate camera gimbals, robotic arms, infrared modules, and other components, facilitating rapid development and customization.

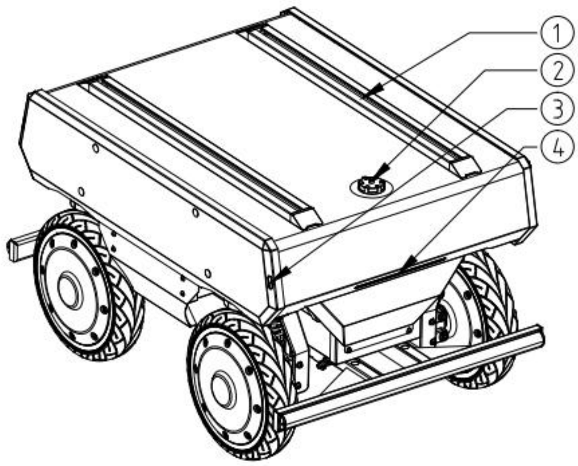



Figure 3 - 1 Tail overall figure

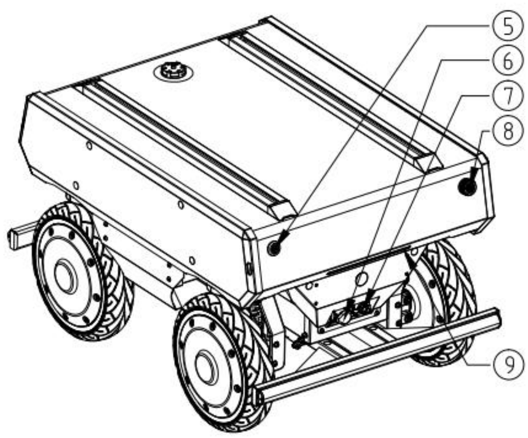



Figure 3 - 2 Front overall figure

Note: ①aluminium profile; ②power supply/comminucation wire port; ③turn signal lights; ④head lightl; ⑤battery power button(internal); ⑥start up button; ⑦debugging/charging quick detach board; ⑧emergancy stop button; ⑨rear tail light/brake light.

## 3.1. Instructions of electrical interface

### 3.1.1. Instructions of tail electrical board

FW-mid pro is equipped with accessible WS32-11 plug-ins power ports on the top. There are 48V, 24V and 12V power supply, with corresponding power supply voltage labels. The red wire of the power supply plug-in is the positive pole, and the black is the negative pole. The pin definition is shown in Table 3-1. The corresponding power supply plug-in wires and communication plug-in wires have been prepared as shown in Figure 3-3, which is convenient for use to provide power for different expansion devices and communication.

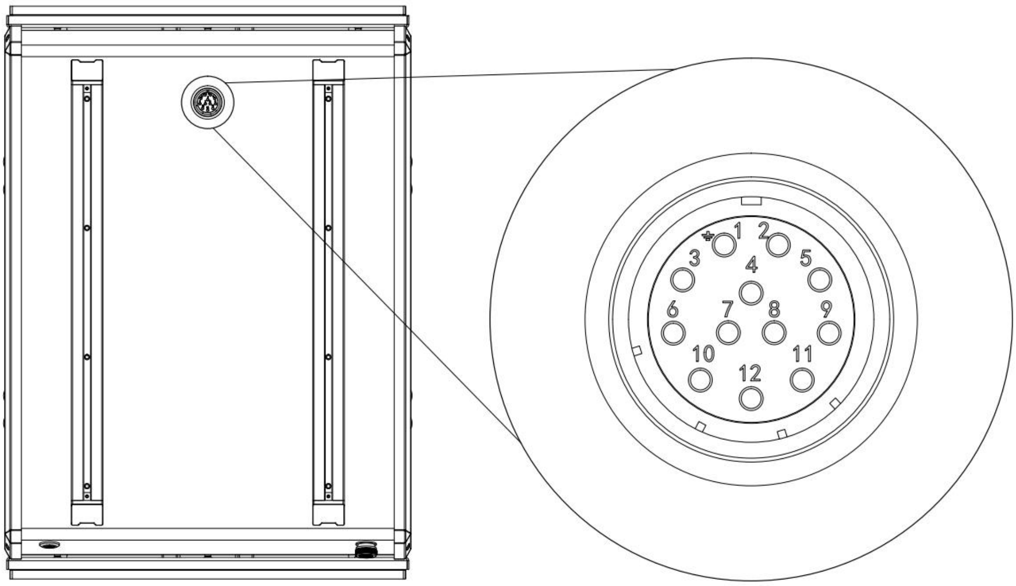



Figure 3 -3 Schematic Diagram of Top Electrical Position

| Pin | Type | Definition | Remark |
|---|---|---|---|
| 1 | CAN | CAN2_H | CAN2 High |
| 4 | | CAN2_L | CAN2 Low |
| 2 | BAT power supply | 48V- | Negative pole of 48V power supply |
| 3 | | 48V+ | Positive pole of 48V power supply |
| 5 | 24V power supply | 24V- | Positive pole of 24V power supply |
| 6 | | 24V+ | Negative pole of 24V power supply |
| 7 | Reserve | NC | NC |

| | | | |
|---|---|---|---|
| 8 | | NC | NC |
| 9 | 12V power supply | 12V- | Positive pole of 12V power supply |
| 10 | | 12V+ | Negative pole of 12V power supply |
| 11 | Ultrasonic sensor CAN | CAN1_H | CAN1 High |
| 12 | | CAN1_L | CAN1 Low |

Table 3-1 Pin Definitions of Top Electrical Interface

## 3.2. Instructions of FW-mid pro remote control

### 3.2.1. FS-i6S remote control operation

Each FW-mid pro is equipped with an FS-i6S remote controller, allowing users to easily control the FW-mid pro. In this product, the remote controller is designed with left-hand control for left and right directions and right-hand control for forward and backward throttle. The definitions and functions can be referred to in Figure 3-4.

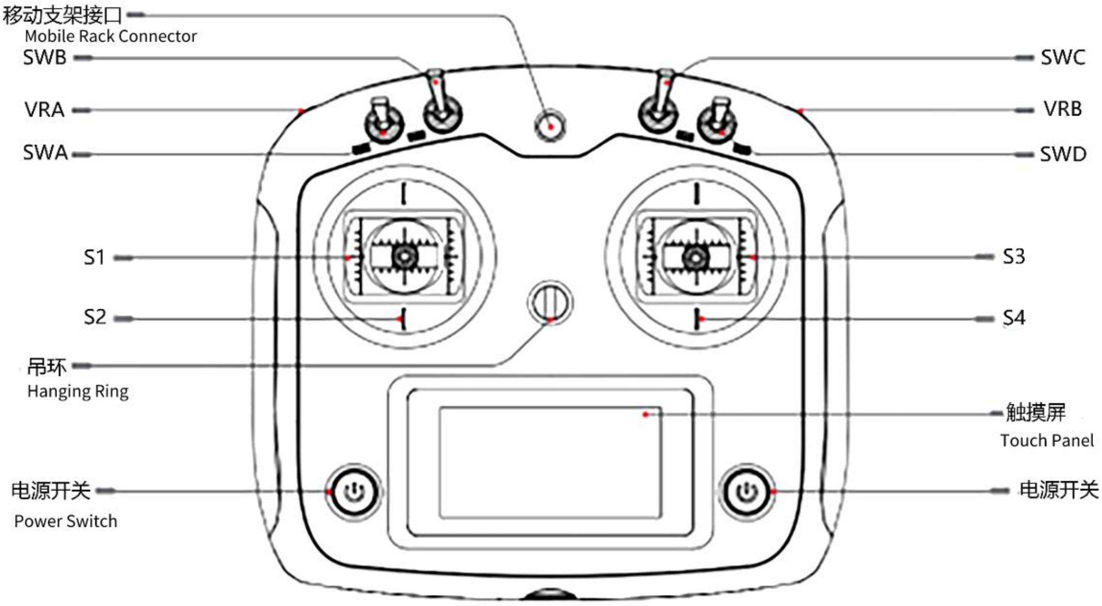



Figure 3-4 Schematic diagram of FS-i6S remote controller keys

The parameters of the remote controller have been configured before delivery. DO NOT modify the system configuration of the remote controller without permission, or, the robot may be out of control and in controlling chaos, etc. In case of any question, please contact the customer services or after-sales personnel for answering:

(1) SWA is the control mode switch lever, with two positions. Taking the remote control facing up as an example: when the SWA lever is in the upper position, it is in the remote control mode, and when the SWA lever is in the lower position, it is in the command control mode.

(2) The SWB is the gear shift lever with three positions. When the lever is moved up, it switches to the parking gear, causing the chassis wheels to enter the parking state, preventing the chassis from moving. When the lever is in the middle position, it switches to the four-wheel drive mode (4T4D), operating in a dual Ackermann steering mode. When the lever is moved down, it switches to the lateral movement gear, where the chassis steering mode allows for lateral movement.

(3) The SWC is the speed switch lever with three positions. When the lever is moved up, it activates the low-speed mode. When the lever is moved down, it activates the high-speed mode. When the lever is in the middle position, it activates the medium-speed mode.

(4) The VRA (Vehicle Release Actuator) is the safety parking unlock dial used to release the safety parking feature. When the collision sensor detects a collision, it triggers the safety parking function. To release the safety parking, you need to turn the VRA dial once (or move the right joystick S4 in the opposite direction of the collision). This action will unlock the safety parking and allow you to continue operating the vehicle.

(5) The VRB is the operational protection dial. When operating the joystick, you need to simultaneously hold down the VRB dial. If you don't hold down the VRB dial, the chassis will not receive any motion commands from the joystick. This serves as a safety measure to prevent unintentional or unauthorized control of the vehicle.

(6) The left joystick is the direction control joystick. Joystick S1 controls the left and right steering of the chassis through its left and right movements. Joystick S2, when moved up or down, does not affect the chassis movement and is not currently in use. Its up and down movements have no impact on the chassis motion.

(7) The right joystick is the throttle control joystick. Joystick S4 controls the forward and backward movement of the chassis through its front and back movements. Joystick S3, when moved left or right, is not currently in use and does not affect the chassis movement. Its left and right movements have no impact on the chassis motion.

(8) There are power buttons on the left and right sides. By simultaneously long-pressing both power buttons, you can perform the power on/off operation.

(9) The standby interface is explained as follows:

The start page is divided into four sections. The top left section displays two timers, T1 and T2. The bottom left section indicates the flight mode. The top right section shows the battery level, with TX representing the remote controller battery and RX representing the robot's battery. The bottom right section includes the unlock button and fine-tuning buttons.

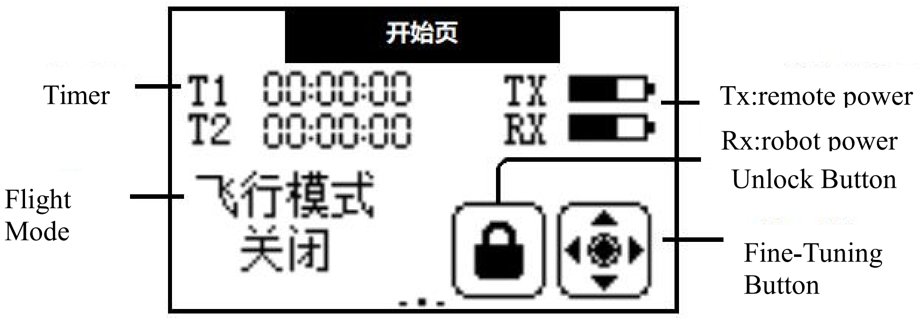



Figure 3-5 Remote controller monitor display interface

The left side of the remote controller's start page is the channel interface, as shown in the figure 3-6:


Figure 3-6 remote controller channel interface

The corresponding relationships between each channel and the remote controller operating components are as table 3-1:

Table 3-1 The corresponding relationships between channel and remote controller

| Channel No. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Remote controller parts | S3 | S2 | S4 | S1 | VRA | VRB | SWA | SWB | SWC | SWD |

The right side of the remote controller's start page is the sensor list page, which includes the following information:

TX.V: Remote Controller Battery Voltage

Int.V: Receiver Voltage

Sig.S: Signal Strength (Normal signal strength is 10)

Ext.V: Robot Chassis Battery Remaining Capacity (Note: the unit show is V, actually should be %, for example, as figure 3-7 show is 66%)

| Name | NO | Value |
|---|---|---|
| TX. V | 0 | 5.20V |
| Int. V | 0 | 4.99V |
| Sig. S | 0 | 10 |
| Ext. V | 1 | 66.00V |

Figure 3-7 The sensor list page of remote controller

- **The control authority regarding the remote controller and communication commands is as follows:**

(1) In scenarios where no remote controller is present: upon startup, the FW-mid pro mobile robot chassis will receive communication commands and execute them accordingly. It will rely solely on these instructions for its operation.

(2) In scenarios where both the remote controller and communication commands are present: the remote controller takes priority in controlling the FW-mid pro mobile robot chassis. It will control the device based on the mode set by the SWA switch on the remote controller. The control authority can be easily obtained by using the SWA switch.

(3) When only the remote controller is present: the control of the FW-mid pro mobile robot chassis is determined by the remote controller's mode, which is controlled by the remote controller itself.

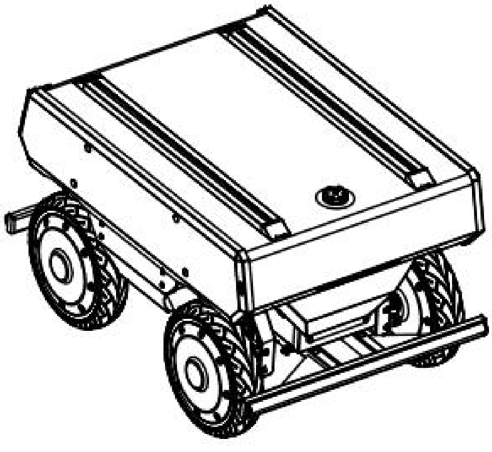
### 3.2.2. Remote control buzzer warning instructions

Table 3-3 Instructions of remote controller alarm condition

| | |
|---|---|
| Switch position alarm | When the remote control is turned on and the lever switches SWA/SWB/SWC/SWD are not in their default positions, an alarm interface will appear, prompting the user to move all the switches to the upward position. Once all the switches are in their default positions, the main interface will appear normally. |
| Low voltage alarm | When the voltage drops below the alarm voltage, the system will emit an alarm, and the remote control screen will start flashing. If the voltage of the remote control is too low, the TX icon will flash, and if the voltage of the chassis is too low, the RX icon will flash. |
| Communication abnormal alarm | When the distance between the remote control and the chassis is too far or there is obstruction interference in the environment, the strength of the remote control signal will decrease. If the signal strength drops below 5, it will trigger a communication abnormal alarm, reminding the user that the remote control signal strength is weak. |
| Remote control unused alarm | When the remote control is unused for a long time, the remote control buzzer will emit intermittent alarms. |
| Power off alarm | When the remote control is turned off, it will check whether the chassis is also turned off. If the chassis is not turned off, a warning interface will pop up, and the chassis power must be turned off before the remote control can be turned off. (If it is necessary to force the remote control to shut down while the chassis is still on, the battery must be removed.) |

### 3.2.3. Instructions of control commands and movement

We will establish a coordinate reference system for ground moving vehicles according to the ISO 8855 standard, as shown in Figure 3-8.



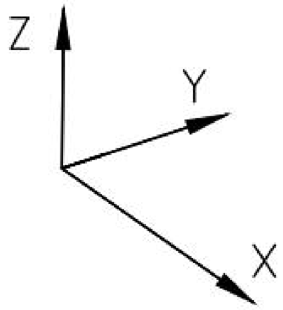


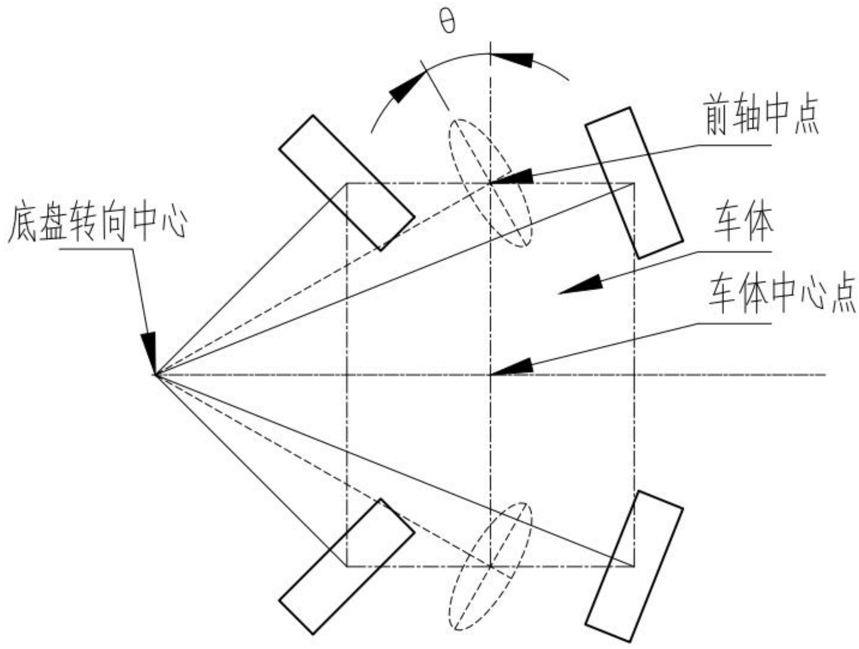

Figure 3-8

As shown in Figure 3-8, the FW-mid pro chassis is parallel to the X-axis of the established reference coordinate system.

In the remote controller control mode, while holding down the VRB (Vehicle Release Actuator) for operational protection, pushing the right-hand throttle joystick S4 forward on the remote controller will move the chassis in the positive X-direction. Pulling it backward will move the chassis in the negative X-direction. When the S4 joystick is pushed to its maximum value, the speed of movement in the X-direction is the highest. When pushed to its minimum value, the speed of movement in the negative X-direction is the highest.

The left-hand direction joystick S1 on the remote controller is used to control the steering movement of the chassis. When the SWB (Shift-Wheel Button) lever is in the four-wheel drive mode (4T4D), pushing the S1 joystick to the left will make the chassis turn left. Pushing it to the maximum left position will result in the highest left turning angular velocity. Likewise, pushing the S1 joystick to the right will make the chassis turn right, and pushing it to the maximum right position will result in the highest right turning angular velocity.

Description of the Virtual Steering Wheel




The schematic diagram is as follows:

- **Vehicle body X-axis linear velocity**, **vehicle body Z-axis angular velocity**, and **vehicle body Y-axis linear velocity** are referenced to the center point of the vehicle body.
- **Virtual steering wheel speed** and **virtual steering wheel angle** are referenced to a virtual steering wheel located at the midpoint of the front axle.

**Special Note:**
This virtual steering wheel corresponds to the front wheel of a bicycle model after simplifying a four-wheel steering, four-wheel drive (4WS4WD) vehicle into a bicycle model.

- **Virtual steering wheel speed** refers to the speed of this wheel.
- **Virtual steering wheel angle** refers to the angle between the centerline of the wheel and the centerline of the vehicle body, i.e., the angle **θ** marked in the diagram.

# 4. Use and Operation

This part mainly introduces the basic operation and use of FW-mid pro platform, and how to
conduct secondary development to the vehicle body through CAN bus protocol.

## 4.1. Use and Operation

### 4.1.1. The basic operations flow of remote operation are as follows:

**Inspection**

(1) Check the status of the vehicle body. Check that whether the vehicle body has obvious
abnormality; If any, please contact after-sales support;

(2) Check the status of the emergency stop button, and confirm that the emergency stop button at the tail is under the released state;

(3) Check that all gears of the remote controller are in default position;

**Start-up**

(1) Press Power Button

(2) Check the battery voltage to see if the battery voltage is normal, if the voltage is too low, please charge it first.

## 4.2. Charging

FW-mid pro mobile robot is equipped with a 48V/10A charger in default, meeting the demands of charging of the users.

The specific operation processes of charging are as follows:

(1) Before charging, make sure that the FW-mid pro is powered off and that the main power switch is off.

(2) First, insert the output plug of the charger into the charging interface on the electrical board at the tail; Then, plug the AC plug of the charger into the 220VAC socket. When the indicator light is red, it enters the charging state, and when the indicator light is green, the charging is completed..

(3) Reverse the process after charging is completed. Unplugging the AC plug first, then the output plug.

(4) The charger protection status description is shown in Table 4- 1:

Table 4 - 1 charger protection status description

| Protective Function | Functional Description |
|---|---|

| | |
|---|---|
| Overheating Protection | When the internal temperature of the charger reaches the over-temperature protection point, the charger will automatically stop charging. |
| Output Short Circuit Protection | The charger will automatically shut off the output when the charger output is accidentally short-circuited. |
| Output Reverse Connection Protection | When the battery is reversed, the charger cuts off the connection between the internal circuit and the battery. |
| Output Over-Voltage Protection | The charger automatically shuts off the output when over-voltage occurs at the charger output. |

ATTENTION:

The charging process must be operated in order to prevent the charger and socket from being electrified and the battery charging port from short-circuiting, resulting in damage to the robot's battery, charger, or unnecessary personal injury.

## 4.3. CAN Wire Connection

FW-mid pro chassis provides CAN interface to users for development, and users can conduct command control to the vehicle with CAN interface by connecting the CAN adapter to the connector on the top electrical port.

## 4.4. CAN Interface Protocol

The communication of FW-mid pro chassis is conducted by CAN2.0B extended frame, and the message format is Intel format with a baud rate of 500K. Through the external CAN bus interface,the vehicle speed, steering angle and steering angular speed of the chassis can be controlled. The FW-mid pro will feed back the current movement state information and the system state information of the chassis in real time.

The specific protocol contents are shown as below:

**The motion command control frame includes gear control, linear speed control, steering angle control, steering angular speed control, light control, parking request and inspection, etc.** The specific protocol contents are shown in Table 4-2.

Table 4-2 Command Control Frame and System Feedback Frame

| Movement control command - control frame | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Message Name** | | | **ID** | | | **Type** | **Cycle (ms)** | | **Message Length (Byte)** |
| ctrl_cmd | | | 0x18C4D1D0 | | | Cycle | 10~50 | | 8 |
| **Signal Descrip-tion** | **For-mat** | **Start-ing Byte** | **Start-ing Bit** | **Signal Trans-mission Type** | **Signal Dura-tion** | **Data Type** | **Preci-sion** | **Unit** | **Signal Value Description** |
| Target gear | Intel | 0 | 0 | Cycle | 4 | Unsigned | 1 | | 00: disable<br>01: Gear Parking<br>02: Gear Neutral<br>06: Gear 4T4D<br>08: Gear Parallel Moving |
| Target vehicle X-axis linear velocity | Intel | 0 | 4 | Cycle | 16 | signed | 0.001 | m/s | 0.001m/s/bit<br><br>Only available under 06: Gear 4T4D or<br>08: Gear Parallel Moving |
| Target vehicle Z-axis angular velocity | Intel | 2 | 20 | Cycle | 16 | signed | 0.01 | °/s | (0.01°/s)/bit<br><br>Only available under 06: Gear 4T4D |
| Target vehicle Y-axis linear velocity | Intel | 4 | 36 | Cycle | 16 | signed | 0.001 | m/s | 0.001m/s/bit<br><br>Only available under<br>08: Gear Parallel Moving |
| Alive Rolling Counter Heartbeat Signal (loop counter) | Intel | 6 | 52 | Cycle | 4 | Unsigned | 1 | | For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnec- |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | tion. |
| Check BCC XOR checkout for message | Intel | 7 | 56 | Cycle | 8 | Unsigned | 1 | | Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6 |

| Movement control command - control frame | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Message Name** | | **ID** | | | | **Type** | | **Cycle (ms)** | **Message Length (Byte)** |
| steering_ctrl_cmd | | 0x18C4D2D0 | | | | Cycle | | 10~50 | 8 |
| **Signal Descrip-tion** | **For-mat** | **Start-ing Byte** | **Start-ing Bit** | **Signal Trans-mission Type** | **Signal Dura-tion** | **Data Type** | **Preci-sion** | **Unit** | **Signal Value Description** |
| Target gear | Intel | 0 | 0 | Cycle | 4 | Unsi-gned | 1 | | 00: disable<br>01: Gear Parking<br>02: Gear Neutral<br>05: Gear 4T4D<br>07: Gear Lateral Moving |
| Target virtual steering wheel's velocity | Intel | 0 | 4 | Cycle | 16 | signed | 0.001 | m/s | 0.001m/s/bit;<br><br>Only available under 05: Gear 4T4D or<br>07: Gear Lateral Moving |
| Target virtual steering wheel's angle | Intel | 2 | 20 | Cycle | 16 | signed | 0.01 | ° | 0.01°/bit;<br><br>Only available under 05: Gear 4T4D or<br>07: Gear Lateral Moving |
| Alive Rolling Counter Heartbeat Signal (loop counter) | Intel | 6 | 52 | Cycle | 4 | Unsigned | 1 | | For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnec-tion. |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Check BCC XOR checkout for message | Intel | 7 | 56 | Cycle | 8 | Unsigned | 1 | | Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6 |

| I/O control command - control frame | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Message Name** | **ID** | | | | | **Type** | | **Cycle (ms)** | **Message Length (Byte)** |
| io_cmd | 0x18C4D7D0 | | | | | IfActive | | 10 | 8 |
| **Signal Descrip-tion** | **For-mat** | **Start-ing Byte** | **Start-ing Bit** | **Signal Trans-mission Type** | **Signal Dura-tion** | **Data Type** | **Preci-sion** | **Unit** | **Signal Value Description** |
| Lamp control | Intel | 0 | 0 | IfAc-tive | 1 | Unsigned | 1 | | 0 = automatic control according to status<br>1 = free control |
| Safety parking unlock switch | Intel | 0 | 1 | IfAc-tive | 1 | Unsigned | 1 | | 0 = invalid<br>1 = unlock enable |
| Low consumption mode switch | Intel | 0 | 1 | IfActive | 1 | Unsigned | 1 | | 0 = off<br>1 = on (default) |
| Lower beam switch | Intel | 1 | 8 | IfAc-tive | 1 | Unsigned | 1 | | 0 = off<br>1 = on |
| Steering lamp switch | Intel | 1 | 10 | IfAc-tive | 2 | Unsigned | 1 | | 0 = off<br>1 = left steering lamp on<br>2 =right steering lamp on |
| Braking lamp switch | Intel | 1 | 12 | IfAc-tive | 1 | Unsigned | 1 | | 0 = off<br>1 = on |
| Low consumption mode triggering ratio | Intel | 2 | 16 | IfActive | 8 | Unsigned | 1 | | 1%/bit;<br>Invalid when less than 0 or greater than 100;<br>The higher the ratio, the harder it is to trigger. |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Alive Rolling Counter Heartbeat Signal(loop counter) | Intel | 6 | 52 | Cy-cle | 4 | Unsigned | 1 | | For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection. |
| Check BCC XOR checkout for message | Intel | 7 | 56 | Cy-cle | 8 | Unsigned | 1 | | Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6 |

| Movement control status - feedback frame | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Message Name | | | Msg ID | | | Type | Cycle (ms) | Message Length (Byte) | |
| ctrl_fb | | | 0x18C4D1EF | | | Cycle | 10 | 8 | |
| **Signal Descrip-tion** | **For-mat** | **Start-ing Byte** | **Start-ing Bit** | **Signal Trans-mission Type** | **Signal Dura-tion** | **Data Type** | **Preci-sion** | **Unit** | **Signal Value Description** |
| Current gear feedback | Intel | 0 | 0 | Cycle | 4 | Unsigned | 1 | | 00: disable<br>01: Gear Parking<br>02: Gear Neutral<br>06: Gear 4T4D<br>08: Gear Lateral Moving |
| Current linear X-axis linear velocity feedback | Intel | 0 | 4 | Cycle | 16 | signed | 0.001 | m/s | 0.001m/s/bit |
| Current Z-axis angular velocity feedback | Intel | 2 | 20 | Cycle | 16 | signed | 0.01 | °/s | (0.01°/s)/bit |
| Current Y-axis linear velocity feedback | Intel | 4 | 36 | Cycle | 16 | signed | 0.001 | m/s | 0.001m/s/bit |
| Alive Rolling Counter Heartbeat (loop counter) | Intel | 6 | 52 | Cycle | 4 | Unsigned | 1 | | For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnec-tion. |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Check BCC XOR checkout for message | Intel | 7 | 56 | Cycle | 8 | Unsigned | 1 | | Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6 |

| Movement control status - feedback frame | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Message Name | | | Msg ID | | Type | Cycle (ms) | | Message Length (Byte) | |
| steering_ctrl_fb | | | 0x18C4D2EF | | Cycle | 10 | | 8 | |
| Signal Descrip-tion | For-mat | Start-ing Byte | Start-ing Bit | Signal Trans-mission Type | Signal Dura-tion | Data Type | Preci-sion | Unit | Signal Value Description |
| Current gear feedback | Intel | 0 | 0 | Cycle | 4 | Unsi-gned | 1 | | 00： disable<br>01： Gear Parking<br>02： Gear Neutral<br>05： Gear 4T4D<br>07： Gear Lateral Moving |
| Current virtual steering wheel's velocity feedback | Intel | 0 | 4 | Cycle | 16 | signed | 0.001 | m/s | 0.001m/s/bit； |
| Current virtual steering wheel's angle feedback | Intel | 2 | 20 | Cycle | 16 | signed | 0.01 | °/s | (0.01°/s)/bit； |
| Alive Rolling Counter Heartbeat Signal(loop counter) | Intel | 6 | 52 | Cycle | 4 | Unsigned | 1 | | For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection |
| Check BCC XOR checkout for message | Intel | 7 | 56 | Cycle | 8 | Unsigned | 1 | | Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6 |

| Wheels control status - feedback frame | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Message Name** | | | **Msg ID** | | | **Type** | **Cycle (ms)** | | **Message Length (Byte)** |
| lf_wheel_fb | | | 0x18C4D6EF | | | Cycle | 10 | | 8 |
| **Signal Descrip -tion** | **For-mat** | **Start-ing Byte** | **Start -ing Bit** | **Signal Trans-mission Type** | **Signal Dura -tion** | **Data Type** | **Preci -sion** | **Unit** | **Signal Value Description** |
| Current left front wheel velocity feedback | Intel | 0 | 0 | Cycle | 16 | signed | 0.001 | m/s | 0.001m/s/bit; |
| Current left front wheel pulse feedback | Intel | 2 | 16 | Cycle | 32 | signed | 1 | 1 | N pulses for single wheel per turn,N = encoder lines 4096 * reduction ratio1 |
| Alive Rolling Counter Heartbeat Signal(loop counter) | Intel | 6 | 52 | Cycle | 4 | Unsigned | 1 | | For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection. |
| Check BCC XOR checkout for message | Intel | 7 | 56 | Cycle | 8 | Unsigned | 1 | | Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6 |

| Wheels control status - feedback frame | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Message Name | | Msg ID | | | Type | Cycle (ms) | | Message Length (Byte) | |
| lr_wheel_fb | | 0x18C4D7EF | | | Cycle | 10 | | 8 | |
| Signal Descrip -tion | For-mat | Start -ing Byte | Start -ing Bit | Signal Trans - mission Type | Signal Dura -tion | Data Type | Precision | Unit | Signal Value Description |
| Current left rear wheel velocity feedback | Intel | 0 | 0 | Cycle | 16 | signed | 0.001 | m/s | 0.001m/s/bit; |
| Current left rear wheel pulse feedback | Intel | 2 | 16 | Cycle | 32 | signed | 1 | 1 | N pulses for single wheel per turn,N = encoder lines 4096 * reduction ratio1 |
| Alive Rolling Counter Heartbeat Signal(loop counter) | Intel | 6 | 52 | Cycle | 4 | Unsi -gned | 1 | | For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection. |
| Check BCC XOR checkout for message | Intel | 7 | 56 | Cycle | 8 | Uns -igned | 1 | | Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6 |

| Wheels control status - feedback frame | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Message Name** | | **Msg ID** | | | **Type** | **Cycle (ms)** | | | **Message Length (Byte)** |
| rr_wheel_fb | | 0x18C4D8EF | | | Cycle | 10 | | | 8 |
| **Signal Descrip-tion** | **For-mat** | **Start-ing Byte** | **Start -ing Bit** | **Signal Trans-mission Type** | **Signal Dura-tion** | **Data Type** | **Preci-sion** | **Unit** | **Signal Value Description** |
| Current right rear wheel velocity feedback | Intel | 0 | 0 | Cycle | 16 | signed | 0.001 | m/s | 0.001m/s/bit; |
| Current right rear wheel pulse feedback | Intel | 2 | 16 | Cycle | 32 | signed | 1 | 1 | N pulses for single wheel per turn,N = encoder lines 4096 * reduction ratio1 |
| Alive Rolling Counter Heartbeat Signal(loop counter) | Intel | 6 | 52 | Cycle | 4 | Unsigned | 1 | | For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection. |
| Check BCC XOR checkout for message | Intel | 7 | 56 | Cycle | 8 | Unsigned | 1 | | Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6 |

| Wheels control status - feedback frame | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Message Name** | | **Msg ID** | | | **Type** | **Cycle (ms)** | | **Message Length (Byte)** | |
| rf_wheel_fb | | 0x18C4D9EF | | | Cycle | 10 | | 8 | |
| **Signal Descrip -tion** | **For-mat** | **Start-ing Byte** | **Start -ing Bit** | **Signal Trans-mission Type** | **Signal Dura -tion** | **Data Type** | **Preci-sion** | **Unit** | **Signal Value Description** |
| Current right front wheel velocity feedback | Intel | 0 | 0 | Cycle | 16 | signed | 0.001 | m/s | 0.001m/s/bit; |
| Current right front wheel pulse feedback | Intel | 2 | 16 | Cycle | 32 | signed | 1 | 1 | N pulses for single wheel per turn,N = encoder lines 4096 * reduction ratio1 |
| Alive Rolling Counter Heartbeat Signal(loop counter) | Intel | 6 | 52 | Cycle | 4 | Unsigned | 1 | | For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection. |
| Check BCC XOR checkout for message | Intel | 7 | 56 | Cycle | 8 | Unsigned | 1 | | Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6 |

| Wheels control status - feedback frame | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Message Name** | | **Msg ID** | | | **Type** | **Cycle (ms)** | | | **Message Length (Byte)** |
| front_angle_fb | | 0x18C4DCEF | | | Cycle | 10 | | | 8 |
| **Signal Descrip-tion** | **For-mat** | **Start-ing Byte** | **Start-ing Bit** | **Signal Trans-mission Type** | **Signal Dura-tion** | **Data Type** | **Preci-sion** | **Unit** | **Signal Value Description** |
| Front left servo joint angle feedback | Intel | 0 | 0 | Cycle | 16 | signed | 0.01 | ° | 0.01°/bit; |
| Front right servo joint angle feedback | Intel | 2 | 16 | Cycle | 16 | signed | 0.01 | ° | 0.01°/bit; |
| Alive Rolling Counter Heartbeat Signal(loop counter) | Intel | 6 | 52 | Cycle | 4 | Unsigned | 1 | | For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection. |
| Check BCC XOR checkout for message | Intel | 7 | 56 | Cycle | 8 | Unsigned | 1 | | Checksum = Byte0 XOR Bye3 XOR te1 XOR Byte2 XOR BytByte4 XOR Byte5 XOR Byte6 |

| Wheels control status - feedback frame | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Message Name** | | | **Msg ID** | | | **Type** | **Cycle（ms）** | | **Message Length (Byte)** |
| rear_angle_fb | | | 0x18C4DDEF | | | Cycle | 10 | | 8 |
| **Signal Descrip-tion** | **For-mat** | **Start-ing Byte** | **Start-ing Bit** | **Signal Trans-mission Type** | **Signal Dura-tion** | **Data Type** | **Preci-sion** | **Unit** | **Signal Value Description** |
| Rear left servo joint angle feedback | Intel | 0 | 0 | Cycle | 16 | signed | 0.01 | ° | 0.01°/bit; |
| Rear right servo joint angle feedback | Intel | 2 | 16 | Cycle | 16 | signed | 0.01 | ° | 0.01°/bit; |
| Alive Rolling Counter Heartbeat Signal(loop counter) | Intel | 6 | 52 | Cycle | 4 | Unsigned | 1 | | For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection. |
| Check BCC XOR checkout for message | Intel | 7 | 56 | Cycle | 8 | Unsigned | 1 | | Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6 |

| I/O control command - feedback frame | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Message Name** | | **Msg ID** | | | **Type** | **Cycle (ms)** | | **Message Length (Byte)** |
| io_fb | | 0x18C4DAEF | | | Cycle | 50 | | 8 |
| **Signal Description** | **Format** | **Starting Byte** | **Starting Bit** | **Signal Transmission Type** | **Signal Duration** | **Data Type** | **Precision** | **Unit** | **Signal Value Description** |
| Light control status feedback | Intel | 0 | 0 | Cycle | 1 | Unsigned | 1 | | 0 = automatic control according to status 1 = free control |
| Unlock status feedback | Intel | 0 | 1 | IfActive | 1 | Unsigned | 1 | | 0 = invalid<br>1 = Unlock succeeded (only prompt once) |
| Low consumption mode status feedback | Intel | 0 | 2 | Cycle | 1 | Unsigned | 1 | | 0 = off<br>1 = on |
| Low consumption mode triggering status feedback | Intel | 0 | 3 | Cycle | 1 | Unsigned | 1 | | 0 = not triggered<br>1 = triggered |
| Lower beam switch on/off status feedback | Intel | 1 | 8 | Cycle | 1 | Unsigned | 1 | | 0 = off<br>1 = on |
| Steering lamp power on/off status feedback | Intel | 1 | 10 | Cycle | 2 | Unsigned | 1 | | 0 = off<br>1 = left steering lamp on<br>2 =right steering lamp on<br>3 =left and right steering lamp on |
| Braking lamp power on/off status feedback | Intel | 1 | 12 | Cycle | 1 | Unsigned | 1 | | 0 = off<br>1 = on |
| Low consumption triggering ratio | Intel | 2 | 16 | Cycle | 8 | Unsigned | 1 | | 1%/bit |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| feedback | | | | | | | | | |
| Front left bumper status feedback | Intel | 1 | 24 | Cycle | 1 | Unsigned | 1 | | 0 = not triggered<br>1 = triggered |
| Front middle bumper status feedback | Intel | 1 | 25 | Cycle | 1 | Unsigned | 1 | | 0 = not triggered<br>1 = triggered |
| Front right bumper status feedback | Intel | 1 | 26 | Cycle | 1 | Unsigned | 1 | | 0 = not triggered<br>1 = triggered |
| Rear left bumper status feedback | Intel | 1 | 27 | Cycle | 1 | Unsigned | 1 | | 0 = not triggered<br>1 = triggered |
| Rear middle bumper status feedback | Intel | 1 | 28 | Cycle | 1 | Unsigned | 1 | | 0 = not triggered<br>1 = triggered |
| Rear right bumper status feedback | Intel | 1 | 29 | Cycle | 1 | Unsigned | 1 | | 0 = not triggered<br>1 = triggered |
| Emergency stop status feedback | Intel | 5 | 40 | Cycle | 1 | Unsigned | 1 | | 0 = off<br>1 = on |
| Remote controller status feedback | Intel | 5 | 41 | Cycle | 1 | Unsigned | 1 | | 0 = command control status<br>1 = remote controller control status |
| Charging station status feedback | Intel | 5 | 42 | Cycle | 1 | Unsigned | 1 | | 0 = no charging station connect<br>1 = charging station connect |
| Manual charging adapter signal feedback | Intel | 5 | 43 | Cycle | 1 | Unsigned | 1 | | 0 = no charging adapter connect<br>1 = charging adapter connect |
| Remote controller first power-on feedback | Intel | 5 | 44 | Cycle | 1 | Unsigned | 1 | | 0 = not powered on<br>1 = powered on |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Remote controller signal status feedback | Intel | 5 | 45 | Cycle | 1 | Unsigned | 1 | | 0 = not powered on or signal disconnected<br><br>1 = signal connected |
| Alive Rolling Counter Heartbeat Signal(loop counter) | Intel | 6 | 52 | Cycle | 4 | Unsigned | 1 | | For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection. |
| Check BCC XOR checkout for message | Intel | 7 | 56 | Cycle | 8 | Unsigned | 1 | | Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6 |

| Battery status - feedback frame | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Message Name** | | | **Msg ID** | | | **Type** | **Cycle(ms)** | | **Message Length (Byte)** |
| bms_fb | | | 0x18C4E1EF | | | Cycle | 100 | | 8 |
| **Signal Descrip -tion** | **For-mat** | **Start -ing Byte** | **Start -ing Bit** | **Signal Trans-mission Type** | **Signal Dura -tion** | **Data Type** | **Preci-sion** | **Unit** | **Signal Value Description** |
| Battery voltage | Intel | 0 | 0 | Cycle | 16 | Unsigned | 0.01 | V | 0.01V/bit; |
| Battery current | Intel | 2 | 16 | Cycle | 16 | signed | 0.01 | A | 0.01A/bit; |
| Battery remaining capacity | Intel | 4 | 32 | Cycle | 16 | Unsigned | 0.01 | Ah | 0.01Ah/bit; |
| Alive Rolling Counter Heartbeat Signal(loop counter) | Intel | 6 | 52 | Cycle | 4 | Unsigned | 1 | | For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection. |
| Check BCC XOR checkout for message | Intel | 7 | 56 | Cycle | 8 | Unsigned | 1 | | Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6 |

| Battery flag bit status - feedback frame | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Message Name** | | | **Msg ID** | | | **Type** | **Cycle (ms)** | | **Message Length (Byte)** |
| bms_flag_fb | | | 0x18C4E2EF | | | Cycle | 100 | | 8 |
| **Signal Description** | **For-mat** | **Start-ing Byte** | **Start-ing Bit** | **Signal Trans-mission Type** | **Signal Dura-tion** | **Data Type** | **Preci-sion** | **Unit** | **Signal Value Description** |
| Battery SOC | Intel | 0 | 0 | Cycle | 8 | Unsigned | 1 | % | 1%/bit; Unit % |
| Single cell overvoltage protection | Intel | 1 | 8 | Cycle | 1 | Unsigned | 1 | | 0 = off<br>1 = on |
| Single cell undervoltage protection | Intel | 1 | 9 | Cycle | 1 | Unsigned | 1 | | 0 = off<br>1 = on |
| Overvoltage protection for the entire battery pack | Intel | 1 | 10 | Cycle | 1 | Unsigned | 1 | | 0 = off<br>1 = on |
| Undervoltage protection for the entire battery pack | Intel | 1 | 11 | Cycle | 1 | Unsigned | 1 | | 0 = off<br>1 = on |
| Charging over temperature protection | Intel | 1 | 12 | Cycle | 1 | Unsigned | 1 | | 0 = off<br>1 = on |
| Charging under temperature protection | Intel | 1 | 13 | Cycle | 1 | Unsigned | 1 | | 0 = off<br>1 = on |
| Discharging over temperature protection | Intel | 1 | 14 | Cycle | 1 | Unsigned | 1 | | 0 = off<br>1 = on |
| Discharging under temperature | Intel | 1 | 15 | Cycle | 1 | Unsigned | 1 | | 0 = off<br>1 = on |
| Charging over current protection | Intel | 2 | 16 | Cycle | 1 | Unsigned | 1 | | 0 = off<br>1 = on |
| Discharging over current protection | Intel | 2 | 17 | Cycle | 1 | Unsigned | 1 | | 0 = off<br>1 = on |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Short circuit protection | Intel | 2 | 18 | Cycle | 1 | Unsigned | 1 | | 0 = off<br>1 = on |
| Fore-end IC error detection | Intel | 2 | 19 | Cycle | 1 | Unsigned | 1 | | 0 = off<br>1 = on |
| Software locks up MOS | Intel | 2 | 20 | Cycle | 1 | Unsigned | 1 | | 0 = off<br>1 = on |
| Charging flag bit | Intel | 2 | 21 | Cycle | 1 | Unsigned | 1 | | 0 = discharge<br>1 = charge |
| Heater flag bit | Intel | 2 | 22 | Cycle | 1 | Unsigned | 1 | | 0 = not activated<br>1 = activate heating |
| The highest temperature of battery | Intel | 3 | 28 | Cycle | 12 | signed | 0.1 | ℃ | 0.1℃/bit; |
| The lowest temperature of battery | Intel | 5 | 40 | Cycle | 12 | signed | 0.1 | ℃ | 0.1℃/bit; |
| Alive Rolling Counter Heartbeat Signal(loop counter) | Intel | 6 | 52 | Cycle | 4 | Unsigned | 1 | | For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection. |
| Check BCC XOR checkout for message | Intel | 7 | 56 | Cycle | 8 | Unsigned | 1 | | Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6 |

| Driving motor current - feedback frame | | | | |
|---|---|---|---|---|
| **Message Name** | **Msg ID** | **Type** | **Cycle (ms)** | **Message Length (Byte)** |
| drive_motor_current_fb | 0x18C4E3EF | Cycle | 100 | 8 |

| Signal Description | For-mat | Starting Byte | Starting Bit | Signal Trans-mission Type | Signal Dura-tion | Data Type | Preci-sion | Unit | Signal Value Description |
|---|---|---|---|---|---|---|---|---|---|
| Left front drive motor current value | Intel | 0 | 0 | Cycle | 12 | Signed | 0.1 | A | 0.1A/bit |
| Left rear drive motor current value | Intel | 1 | 12 | Cycle | 12 | Signed | 0.1 | A | 0.1A/bit |
| Right front drive motor current value | Intel | 3 | 24 | Cycle | 12 | Signed | 0.1 | A | 0.1A/bit |
| Right rear drive motor current value | Intel | 4 | 36 | Cycle | 12 | Signed | 0.1 | A | 0.1A/bit |
| Left front drive motor stall flag | Intel | 6 | 48 | Cycle | 1 | Unsigned | 1 | | 0 = no<br>1 = yes |
| Left rear drive motor stall flag | Intel | 6 | 49 | Cycle | 1 | Unsigned | 1 | | 0 = no<br>1 = yes |
| Right front drive motor stall flag | Intel | 6 | 50 | Cycle | 1 | Unsigned | 1 | | 0 = no<br>1 = yes |
| Right rear drive motor stall flag | Intel | 6 | 51 | Cycle | 1 | Unsigned | 1 | | 0 = no<br>1 = yes |
| Alive Rolling Counter Heartbeat Signal(loop counter) | Intel | 6 | 52 | Cycle | 4 | Unsigned | 1 | | For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection. |
| Check BCC XOR checkout for message | Intel | 7 | 56 | Cycle | 8 | Unsigned | 1 | | Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6 |

| Steering motor current - feedback frame | | | | |
|---|---|---|---|---|
| **Message Name** | **Msg ID** | **Type** | **Cycle (ms)** | **Message Length (Byte)** |
| steering_motor_current_fb | 0x18C4E4EF | Cycle | 100 | 8 |

| Signal Description | For-mat | Start-ing Byte | Start-ing Bit | Signal Trans-mission Type | Signal Dura-tion | Data Type | Preci-sion | Unit | Signal Value Description |
|---|---|---|---|---|---|---|---|---|---|
| Left front steering motor current value | Intel | 0 | 0 | Cycle | 12 | Signed | 0.1 | A | 0.1A/bit |
| Left rear steering motor current value | Intel | 1 | 12 | Cycle | 12 | Signed | 0.1 | A | 0.1A/bit |
| Right front steering motor current value | Intel | 3 | 24 | Cycle | 12 | Signed | 0.1 | A | 0.1A/bit |
| Right rear steering motor current value | Intel | 4 | 36 | Cycle | 12 | Signed | 0.1 | A | 0.1A/bit |
| Left front steering motor stall flag | Intel | 6 | 48 | Cycle | 1 | Unsigned | 1 | | 0 = no<br>1 = yes |
| Left rear steering motor stall flag | Intel | 6 | 49 | Cycle | 1 | Unsigned | 1 | | 0 = no<br>1 = yes |
| Right front steering motor stall flag | Intel | 6 | 50 | Cycle | 1 | Unsigned | 1 | | 0 = no<br>1 = yes |
| Right rear steering motor stall flag | Intel | 6 | 51 | Cycle | 1 | Unsigned | 1 | | 0 = no<br>1 = yes |
| Alive Rolling Counter Heartbeat Signal(loop counter) | Intel | 6 | 52 | Cycle | 4 | Unsigned | 1 | | For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection. |
| Check BCC XOR checkout for message | Intel | 7 | 56 | Cycle | 8 | Unsigned | 1 | | Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6 |

| Front ultrasonic radar - feedback frame(reserve) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Message Name** | | | **Msg ID** | | **Type** | **Cycle (ms)** | | **Message Length (Byte)** | |
| front_ultrasonic_fb | | | 0x18C4E8EF | | Cycle | 50 | | 8 | |
| **Signal Description** | **For-mat** | **Starting Byte** | **Starting Bit** | **Signal Trans-mission Type** | **Signal Dura-tion** | **Data Type** | **Preci-sion** | **Unit** | **Signal Value Description** |
| No.1 ultrasonic radar probe's distance | Intel | 0 | 0 | Cycle | 12 | Unsigned | 1 | mm | 1mm/bit within the detection range; 200mm = blind zone 2540mm = over range 2560mm = no probe detection |
| No.2 ultrasonic radar probe's distance | Intel | 1 | 12 | Cycle | 12 | Unsigned | 1 | mm | 1mm/bit within the detection range; 200mm = blind zone 2540mm = over range 2560mm = no probe detection |
| No.3 ultrasonic radar probe's distance | Intel | 3 | 24 | Cycle | 12 | Unsigned | 1 | mm | 1mm/bit within the detection range; 200mm = blind zone 2540mm = over range 2560mm = no probe detection |
| No.4 ultrasonic radar probe's distance | Intel | 4 | 36 | Cycle | 12 | Unsigned | 1 | mm | 1mm/bit within the detection range; 200mm = blind zone 2540mm = over range 2560mm = no probe detection |
| Alive Rolling Counter Heartbeat Signal(loop counter) | Intel | 6 | 52 | Cycle | 4 | Unsigned | 1 | | For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection. |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Check BCC XOR checkout for message | Intel | 7 | 56 | Cycle | 8 | Unsigned | 1 | | Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6 |

| Rear ultrasonic radar - feedback frame(reserve) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Message Name** | | **Msg ID** | | | **Type** | **Cycle (ms)** | | **Message Length (Byte)** | |
| rear_ultrasonic_fb | | 0x18C4E9EF | | | Cycle | 50 | | 8 | |
| **Signal Description** | **For-mat** | **Start-ing Byte** | **Start-ing Bit** | **Signal Trans-mission Type** | **Signal Dura-tion** | **Data Type** | **Preci-sion** | **Unit** | **Signal Value Description** |
| No.5 ultrasonic radar probe's distance | Intel | 0 | 0 | Cycle | 12 | Unsigned | 1 | mm | 1mm/bit within the detection range; 200mm = blind zone 2540mm = over range 2560mm = no probe detection |
| No.6 ultrasonic radar probe's distance | Intel | 1 | 12 | Cycle | 12 | Unsigned | 1 | mm | 1mm/bit within the detection range; 200mm = blind zone 2540mm = over range 2560mm = no probe detection |
| No.7 ultrasonic radar probe's distance | Intel | 3 | 24 | Cycle | 12 | Unsigned | 1 | mm | 1mm/bit within the detection range; 200mm = blind zone 2540mm = over range 2560mm = no probe detection |
| No.8 ultrasonic radar probe's distance | Intel | 4 | 36 | Cycle | 12 | Unsigned | 1 | mm | 1mm/bit within the detection range; 200mm = blind zone 2540mm = over range 2560mm = no probe detection |
| Alive Rolling Counter Heartbeat Signal(loop counter) | Intel | 6 | 52 | Cycle | 4 | Unsigned | 1 | | For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection. |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Check BCC XOR checkout for message | Intel | 7 | 56 | Cycle | 8 | Unsigned | 1 | | Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6 |

| **Fault Alarm- feedback frame** | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Message Name** | | **Msg ID** | | | | **Type** | **Cycle (ms)** | | **Message Length (Byte)** |
| error_fb | | 0x18C4EAEF | | | | Cycle | 100 | | 8 |
| **Signal Description** | **For-mat** | **Start-ing Byte** | **Start-ing Bit** | **Signal Trans-mission Type** | **Signal Dura-tion** | **Data Type** | **Preci-sion** | **Unit** | **Signal Value Description** |
| Fault level | Intel | 0 | 0 | Cycle | 8 | Unsigned | 1 | | 00: no fault<br>E0: slightly fault<br>E1: general fault<br>E2: serious fault<br>E3: fatal fault |
| Fault components | Intel | 1 | 8 | Cycle | 8 | Unsigned | 1 | | 00: no fault<br>A0: remote controller<br>B0: battery<br>C0: VCU<br>D1: driving system<br>D2: steering system<br>D3: braking system<br>F0: sensors<br>80: others |
| Fault components ID | Intel | 2 | 16 | Cycle | 8 | Unsigned | 1 | | 00: default/no<br>06: front left<br>07: rear left<br>08: front right<br>09: rear right |
| Fault code | Intel | 3 | 24 | Cycle | 8 | Unsigned | 1 | | 0x00: Reset/No abnormality<br>0x1X: General abnormality<br>0x2X: Current |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | abnormality<br>0x3X: Voltage abnormality<br>0x4X: Temperature abnormality<br>0x5X: Hardware abnormality<br>0x6X: Software abnormality<br>0x7X: Accessory abnormality<br>0x8X: Communication abnormality<br>0xF0: Device-specific abnormality<br>(X represents 0,1,2...E,F, etc.)<br>For details, please refer to the table below. |
| Error storage device | Intel | 4 | 32 | Cycle | 16 | Unsigned | 1 | | Please consult with YUHESEN's after-sales engineer. |
| Alive Rolling Counter Heartbeat Signal(loop counter) | Intel | 6 | 52 | Cycle | 4 | Unsigned | 1 | | For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection. |
| Check BCC XOR checkout for message | Intel | 7 | 56 | Cycle | 8 | Unsigned | 1 | | Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6 |

**Fault Code Description Table**

| Code | Description | Code | Description | Code | Description | Code | Description |
|---|---|---|---|---|---|---|---|
| 10 | General abnormality | 20 | Current abnormality | 30 | Voltage abnormality | 40 | Temperature abnormality |
| 11 | Stall | 21 | Overload | 31 | Overvoltage | 41 | Drive over-temperature |
| 12 | Block rotation | 22 | Overcurrent | 32 | Undervoltage | 42 | Motor over-temperature |
| 13 | Speed deviation | 23 | Short circuit | 33 | Voltage limit exceeded | 43 | Power module over-tempera |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | ture |
| 14 | Speed limit exceeded | 24 | Current deviation | 34 | Charging overvoltage | 44 | Battery over-tempera ture |
| 15 | Position deviation | 25 | Current limit exceeded | 35 | Power supply voltage abnormality | 45 | Battery low-temperat ure |
| 16 | Position limit exceeded | 26 | I²T fault | 36 | | 46 | Equipment over-tempera ture |
| 17 | Power limit exceeded | 27 | Charging overcurrent | 37 | | 47 | |
| | | | | | | | |
| **Code** | **Description** | **Code** | **Description** | **Code** | **Description** | **Code** | **Description** |
| 50 | Hardware abnormality | 60 | Software abnormality | 70 | Auxiliary attachment abnormality | 80 | Communicati on abnormality |
| 51 | Encoder fault | 61 | Initialization abnormality | | | 81 | Communicati on hardware error |
| 52 | Hall fault | 62 | | | | 82 | CAN communicatio n overload |
| 53 | EEPROM fault | 63 | | | | 83 | Node guardian error |
| 54 | IGBT fault | 64 | | | | 84 | Protocol error |
| 55 | Braking resistor fault | 65 | | | | 85 | Communicati on timeout |
| 56 | FLASH fault | 66 | | | | 86 | Heartbeat failure |
| 57 | FRAM fault | 67 | | | | 87 | CRC failure |
| 58 | ADC fault | 68 | | | | 88 | Offline fault |
| 59 | Motor phase loss fault | | | | | 89 | Offline alert |
| 5A | Power device faul | | | | | | |
| | | | | | | | |

# 4.5. Instruction of using the CAN communication protocol

## 1. Notes during the test:

1.1 Note that the AliveCounter needs to be sent continuously and cyclically during the sending process.

1.2 Send AliveCounter process, pay special attention to AliveCounter occupies 52 to 55 four bits.

1.3 BYTE [7] check bits for the first 7 Byte checksum: Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6.

1.4 The following routine is a simple control command when using USB CAN, please control the vehicle in accordance with the communication protocol to control the command.

1.5 During the test, please switch the remote control to the command control mode or choose to turn off the remote control.

1.6 Use the computer to connect the CAN card test process due to the possibility of testing the vehicle movement and other conditions, in the test process, please set up the vehicle, such as the vehicle test after the stability of the vehicle test with the procedures for testing when the vehicle is put down.

1.7 Due to the highest priority of the remote control during the landing test, it is advisable to turn on the remote control test, so that it is convenient to switch to the remote control mode at any time during the test.

## 2.Vehicle control command description ctrl_cmd

Vehicle motion control commands are divided into two types: ctrl_cmd message and steering_ctrl_cmd message. Both messages require simultaneous transmission of corresponding commands, heartbeat signals, and XOR checksum to pass protection and validation.

**ctrl_cmd Message**

Controls target vehicle motion via:

Target body X-axis linear velocity

Target body Z-axis angular velocity

Target body Y-axis linear velocity

4T4D gear (06): Only X-axis velocity + Z-axis angular velocity are valid; Y-axis velocity is invalid.

Lateral moving gear (08): Only X-axis velocity + Y-axis velocity are valid; Z-axis angular velocity is invalid.

When target gear is Disable gear / Parking gear / Neutral gear: All three signals are invalid.

**steering_ctrl_cmd Message**

Controls target vehicle motion via:

Target virtual steering wheel speed

Target virtual steering wheel angle

4T4D gear (05) or Lateral moving gear (07): Both virtual steering speed + angle are valid.

When target gear is Disable gear / Parking gear / Neutral gear: Both signals are invalid.

## (1) Control request and feedback of ctrl_cmd message

The target gear signal value occupies 0~3 bits, with a physical range of 00 to 08. The default target gear is 00 (disable gear). The gear assignments are as follows:

01: Parking gear

02: Neutral gear

06: 4T4D gear (four-wheel drive/steer mode)

08: Lateral moving gear

In ctrl_cmd messages, target gears 05 or 07 are invalid

**Example 1: 4T4D Gear (06) Request**

Vehicle mode: Four-wheel steering/drive.

Target X-axis linear velocity:

Scaling: 0.001 m/s per bit * bus signal

Example: To request 0.5 m/s, set bus signal to 500 (0x01F4).

Sign convention: Positive for forward, negative for backward.

Target Z-axis angular velocity:

Scaling: 0.01°/s per bit * bus signal

Example: To request -25°/s, set bus signal to -2500 (0xF63C).

Sign convention: Positive for counterclockwise (top view), negative for clockwise.

Target Y-axis linear velocity: Invalid in this gear; default to 0 (0x0000)

Example commands are as follows:

| ID | D[0] | D[1] | D[2] | D[3] | D[4] | D[5] | D[6] | D[7] |
|---|---|---|---|---|---|---|---|---|
| 0x18C4D1D0 | 0x46 | 0x1F | 0xC0 | 0x63 | 0x0F | 0x00 | 0x10 | 0xE5 |
| 0x18C4D1D0 | 0x46 | 0x1F | 0xC0 | 0x63 | 0x0F | 0x00 | 0x20 | 0xD5 |
| 0x18C4D1D0 | 0x46 | 0x1F | 0xC0 | 0x63 | 0x0F | 0x00 | 0x30 | 0xC5 |

Note: the above three signal frames are cyclically sent at 10ms intervals, enabling the vehicle to operate in four-wheel steering and four-wheel drive mode with an X-axis linear velocity of 0.5 m/s and a Z-axis angular velocity of -25°/s.

Feedback:

| ID | D[0] | D[1] | D[2] | D[3] | D[4] | D[5] | D[6] | D[7] |
|---|---|---|---|---|---|---|---|---|
| 0x18C4D1EF | 0x46 | 0x1F | 0xC0 | 0x63 | 0x0F | 0x00 | 0xA0 | 0x55 |
| 0x18C4D1EF | 0x46 | 0x1F | 0xC0 | 0x63 | 0x0F | 0x00 | 0xB0 | 0x45 |
| 0x18C4D1EF | 0x46 | 0x1F | 0xC0 | 0x63 | 0x0F | 0x00 | 0xC0 | 0x35 |

Note: the XOR checksum and heartbeat signals cycle dynamically, and the feedback value may not be absolute due to automatic speed adjustment during operation.

**Example 2: Lateral Moving Gear (08) Request**

Vehicle mode: Lateral moving mode.

Target X-axis linear velocity:

Scaling: 0.001 m/s per bit * bus signal

Example: To request 0.3 m/s, set bus signal to 300 (0x012C).

Sign convention: Positive for forward, negative for backward.

Target Z-axis angular velocity: Invalid in this gear; default to 0 (0x0000).

Target Y-axis linear velocity:

Scaling: 0.001 m/s per bit * bus signal

Example: To request -0.3 m/s, set bus signal to -300 (0xFED4).

Sign convention: Positive for left, negative for right.

Example commands are as follows:

| ID | D[0] | D[1] | D[2] | D[3] | D[4] | D[5] | D[6] | D[7] |
|---|---|---|---|---|---|---|---|---|
| 0x18C4D1D0 | 0xC8 | 0x12 | 0x00 | 0x00 | 0x40 | 0xED | 0x4F | 0x38 |
| 0x18C4D1D0 | 0xC8 | 0x12 | 0x00 | 0x00 | 0x40 | 0xED | 0x5F | 0x28 |
| 0x18C4D1D0 | 0xC8 | 0x12 | 0x00 | 0x00 | 0x40 | 0xED | 0x6F | 0x18 |

Note: the above three signal frames are cyclically transmitted at 10ms intervals, enabling the vehicle to operate in lateral moving mode with an X-axis linear velocity of 0.3 m/s and a Y-axis linear velocity of -0.3 m/s.

Feedback:

| ID | D[0] | D[1] | D[2] | D[3] | D[4] | D[5] | D[6] | D[7] |
|---|---|---|---|---|---|---|---|---|
| 0x18C4D1EF | 0xC8 | 0x12 | 0x00 | 0x00 | 0x40 | 0xED | 0xDF | 0xA8 |
| 0x18C4D1EF | 0xC8 | 0x12 | 0x00 | 0x00 | 0x40 | 0xED | 0xEF | 0x98 |
| 0x18C4D1EF | 0xC8 | 0x12 | 0x00 | 0x00 | 0x40 | 0xED | 0xFF | 0x88 |

Note: the XOR checksum and heartbeat signals cycle dynamically, and the feedback value may not be absolute due to automatic speed adjustment during operation.

### (2) Control request and feedback of steering_ctrl_cmd message

The target gear signal value occupies 0~3 bits, with a physical range of 00 to 08. The default target gear is 00 (disable gear). The gear assignments are as follows:

01: Parking gear

02: Neutral gear

05: 4T4D gear (four-wheel steering/drive mode)

07: Lateral moving gear

In steering_ctrl_cmd messages, target gears 06 or 08 are invalid.

**Example 1: 4T4D Gear (05) Request**

Vehicle mode: Four-wheel steering/drive mode.

Target virtual steering wheel speed:

Scaling: 0.001 m/s per bit * bus signal

Example: To request 0.3 m/s, set bus signal to 300 (0x012C).

Sign convention: Positive for forward, negative for backward.

Target virtual steering wheel angle:

Scaling: 0.01° per bit * bus signal

Example: To request -45°, set bus signal to -4500 (0xEE6C).

Sign convention: Positive for left, negative for right.

Example commands are as follows:

| ID | D[0] | D[1] | D[2] | D[3] | D[4] | D[5] | D[6] | D[7] |
|---|---|---|---|---|---|---|---|---|
| 0x18C4D1D0 | 0xC5 | 0x12 | 0xC0 | 0xE6 | 0x0E | 0x00 | 0x10 | 0xEF |
| 0x18C4D1D0 | 0xC5 | 0x12 | 0xC0 | 0xE6 | 0x0E | 0x00 | 0x20 | 0xDF |
| 0x18C4D1D0 | 0xC5 | 0x12 | 0xC0 | 0xE6 | 0x0E | 0x00 | 0x30 | 0xCF |

Note: the above three signal frames are cyclically transmitted at 10ms intervals, enabling the vehicle to operate in four-wheel steering and four-wheel drive mode with a steering wheel speed of 0.3 m/s and a steering wheel angle of -45°.

Feedback：

| ID | D[0] | D[1] | D[2] | D[3] | D[4] | D[5] | D[6] | D[7] |
|---|---|---|---|---|---|---|---|---|
| 0x18C4D1EF | 0xC5 | 0x12 | 0xC0 | 0xE6 | 0x0E | 0x00 | 0xA0 | 0x5F |
| 0x18C4D1EF | 0x45 | 0x06 | 0xC0 | 0xE6 | 0x0E | 0x00 | 0xB0 | 0x4F |
| 0x18C4D1EF | 0x45 | 0x06 | 0xC0 | 0xE6 | 0x0E | 0x00 | 0xC0 | 0x3F |

Note: the XOR checksum and heartbeat signals cycle dynamically, and the feedback value may not be absolute due to automatic speed adjustment during operation.

**Example 2: Lateral Moving Gear (07) Request**

Vehicle mode: Lateral moving mode.

Target virtual steering wheel speed:

Scaling: 0.001 m/s per bit * bus signal

Example: To request -0.5 m/s, set bus signal to -500 (0xFE0C).

Sign convention: Positive for forward, negative for backward.

Target virtual steering wheel angle:

Scaling: 0.01° per bit * bus signal

Example: To request 30°, set bus signal to 3000 (0x0BB8).

Sign convention: Positive for left, negative for right.

Example commands are as follows:

| ID | D[0] | D[1] | D[2] | D[3] | D[4] | D[5] | D[6] | D[7] |
|---|---|---|---|---|---|---|---|---|
| 0x18C4D1D0 | 0xC7 | 0xE0 | 0x8F | 0xBB | 0x00 | 0x00 | 0x40 | 0x53 |
| 0x18C4D1D0 | 0xC7 | 0xE0 | 0x8F | 0xBB | 0x00 | 0x00 | 0x50 | 0x43 |
| 0x18C4D1D0 | 0xC7 | 0xE0 | 0x8F | 0xBB | 0x00 | 0x00 | 0x60 | 0x73 |

Note: the above three signal frames are cyclically transmitted at 10ms intervals, enabling the vehicle to operate in transverse mode with a steering wheel speed of -0.5 m/s and a steering wheel angle of 30°.

Feedback:

| ID | D[0] | D[1] | D[2] | D[3] | D[4] | D[5] | D[6] | D[7] |
|---|---|---|---|---|---|---|---|---|
| 0x18C4D1EF | 0xC7 | 0xE0 | 0x8F | 0xBB | 0x00 | 0x00 | 0xD0 | 0xC3 |
| 0x18C4D1EF | 0xC7 | 0xE0 | 0x8F | 0xBB | 0x00 | 0x00 | 0xE0 | 0xF3 |
| 0x18C4D1EF | 0xC7 | 0xE0 | 0x8F | 0xBB | 0x00 | 0x00 | 0xF0 | 0xE3 |

Note: the XOR checksum and heartbeat signals cycle dynamically, and the feedback value may not be absolute due to automatic speed adjustment during operation.

## 3. Explanations of auxiliary control commands

Take the safe parking unlock switch as an example to illustrate, IO port enable control needs to send enable flag bit, heartbeat signal and check bit at the same time.

Example: io_cmd_unlock safety parking unlock switch 0x02

| ID | D[0] | D[1] | D[2] | D[3] | D[4] | D[5] | D[6] | D[7] |
|---|---|---|---|---|---|---|---|---|
| 0x18C4D7D0 | 0x02 | 0x00 | 0X00 | 0x00 | 0x00 | 0x00 | 0x00 | 0x02 |
| 0x18C4D7D0 | 0x02 | 0x00 | 0X00 | 0x00 | 0x00 | 0x00 | 0x10 | 0x12 |
| 0x18C4D7D0 | 0x00 | 0x00 | 0X00 | 0x00 | 0x00 | 0x00 | 0x20 | 0x20 |
| 0x18C4D7D0 | 0x00 | 0x00 | 0X00 | 0x00 | 0x00 | 0x00 | 0x30 | 0x30 |

Note: The above signals (falling edge) are sent down at 20ms intervals to request unlocking of the safety stop switch.

Feedback:

| ID | D[0] | D[1] | D[2] | D[3] | D[4] | D[5] | D[6] | D[7] |
|---|---|---|---|---|---|---|---|---|
| 0x18C4DAEF | 0x00 | 0x00 | 0X00 | 0x00 | 0x00 | 0x00 | 0x00 | 0x02 |
| 0x18C4DAEF | 0x02 | 0x00 | 0X00 | 0x00 | 0x00 | 0x00 | 0x00 | 0x02 |
| 0x18C4DAEF | 0x00 | 0x00 | 0X00 | 0x00 | 0x00 | 0x00 | 0x00 | 0x02 |

Note: Checksums and Alivecounter cycle changes.

# 5. Attention

This section contains some matters to be noted during use, storage and development of FW-mid pro.

## 5.1. Attentions for battery

▲ The battery of FW-mid pro products may not be fully charged when they are delivered. The specific situations CAN be read through FW-mid pro remote controller vehicle chassis voltage display or CAN bus communication interface. As for charging time, when the green indicator is on, indicating that the product has been fully charged;

▲ DO NOT charge the battery after it is exhausted, and please charge in time when the battery voltage is too low;

▲ The working temperature of the battery under discharging is -20°C~50°C, the battery can work normally within the specified temperature range, and the capacity loss is within the error range:

▲ Excessive discharge of the battery is prohibited during use to avoid damage to the battery;

▲ Avoid excessive impact on the battery; the impact beyond the specification may damage the battery, which may lead to battery leakage, heat, smoke, fire or explosion;

▲ In case of obvious battery abnormalities, please stop using the battery immediately!

## 5.2. Attentions for charging

▲ Charging can only be conducted by the charger matching with the battery. DO NOT use the non-original battery, power supply or charger;

▲ Charging can only be conducted under 0°C~45°C. Charging out of this temperature range will lead to battery leakage, heating or serious damage, which may lead to deterioration of battery performance and life;

▲ During charging, if the charger or battery is abnormal or damaged, please remove the charger input line and output line immediately;

▲ If charging cannot be completed within the specified time, please stop charging immediately. Or, the battery may heat, have smoke or get on fire (or explode);

▲ It is not allowed charge the battery of the vehicle body in thunderstorm weather;

▲ It not allowed to charge the battery of the vehicle body in the place which is wet or with rain;

▲ It is not allowed to charge the battery of the vehicle body with high temperature, such as heat source or direct sunlight, etc.;

▲ Charging shall be conducted in the place which is ventilated and without dust;

▲ During charging, it is not allowed to block the air inlet and outlet of the charger, there shall be a space of 10cm at least;

▲ You should follow the charging process in the specified order to prevent the charging plug of the charger from being live when connecting and disconnecting it with the battery charging port. This will help avoid short circuits, potential damage to the robot's battery and charger, as well as any unnecessary personal injuries.

## 5.3. Attentions for usage environment

▲ The working temperature of FW-mid pro is -20℃~50℃, DO NOT use in the environment with the temperature of lower than -20℃ or higher than 50℃;

▲ The best storage temperature for FW-mid pro is 0℃~40℃;

▲ DO NOT store or user in the environment with corrosive, inflammable and explosive gas;

▲ During use and storage, please keep away from heat resources and fire resources;

▲ Excepting for special edition (with customized IP protection level), the water-proof function of FW-mid pro is limited. DO NOT use FW-mid pro in the environment with deep ponding;

## 5.4. Attentions for remote operation

▲ When operating the chassis in remote control mode, it is necessary to hold down the VRB safety switch. Releasing the VRB will automatically stop the chassis and it will no longer respond to the controls of the left-hand direction and right-hand gas pedal rocker.

▲ The emergency stop switch is released; The throttle remote lever returns to zero, that is, the chassis speed is 0;

▲ When controlling the vehicle to move forward using the S4 joystick, if you need to perform a backward operation, you should first return the S4 joystick to the neutral position before engaging the reverse gear. It is prohibited to quickly move the joystick to the reverse position without returning it to neutral. The same principle applies to left and right turns. When changing directions, always return the joystick to neutral before changing.

▲ DO NOT turn off the power of the remote controller while the vehicle is in normal operation. If the battery of the remote controller runs out, the communication will be interrupted, and the protective program will activate, causing the chassis to stop moving after 3 seconds. Once the remote controller is powered on again, communication will be automatically restored, and normal operation can resume.

## 5.5. Attentions for external electrical extension

▲ The top power supply current shall be the battery voltage and current strictly selected. Over-current is not allowed;

▲ When the system detects that the battery voltage is lower than the safe voltage, protection procedure will be started automatically. If the external extension equipment involves storage of important data, and there is not automatic storage function for powering off, please charge timely.

## 5.6. Other attentions

▲ During handling or setting, DO NOT fall or invert;

▲ In case of no professionals, DO NOT disassemble without permission;

▲ If the remote controller end will not be used for a long time, the battery shall be removed;

▲ The tires shall be replaced timely in accordance with the wearing conditions of the patterns on the wheel tread.

▲ When making a customs consignment, press the button on the inside of the left hole in the edge of the skirt with a thin lever tool to turn off the main battery switch.

# 6. Common Q&A

**Q: FW-mid pro starts normally, however, the vehicle body does not move under the control of the remote controller?**

**A**: First, make sure that the emergency stop switch at the rear is released.Next, ensure that the SWA joystick is in remote control mode and the SWB gear is not in the parking position. Confirm that you are holding down the VRB safety switch during the operation. Finally, make sure that you are not charging through the charging port.

**Q: FW-mid pro What should I do that the battery of the remote controller is low, and the vehicle body stops running?**

**A**: Please replace the battery of the remote control immediately, after that, normal communication will recover soon.

**Q: The charger led (green) is off**

**A**: Please firstly check that whether the connection of the charger input interface is correct and firm; And then, check that whether there is AC input.

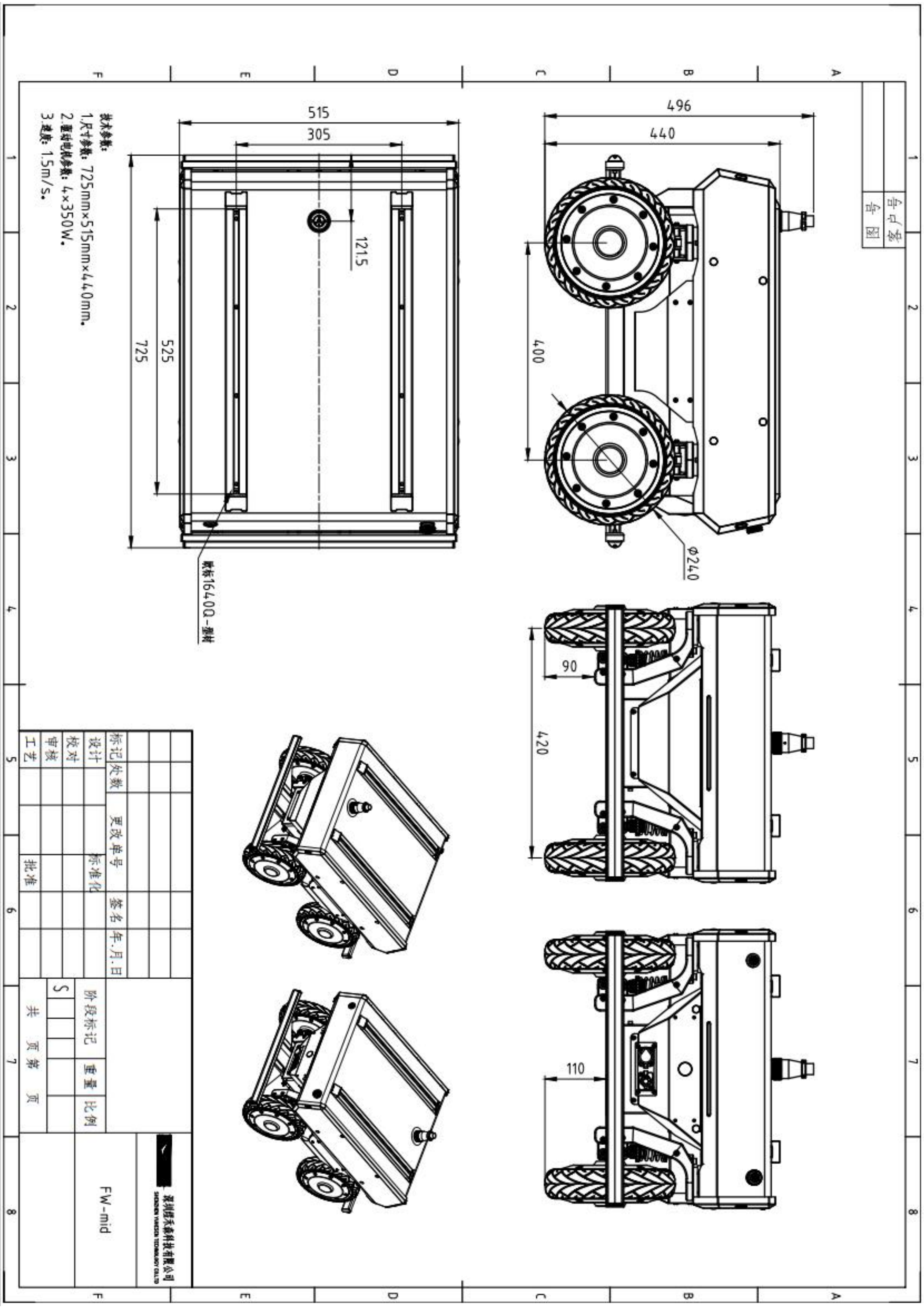
**Q: The charger led (red) is off**

**A**: Please firstly check that whether the connection of the charger input interface is correct and firm;

Whether the battery has not been used for a long time, and whether the battery over-discharges or is damaged;

Re-plug the plugger of input and output line with a time interval of larger than 10s to judge that whether the charger is being protected.

# 7. Drawing



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